

Transformer-Based Parameter Estimation for Metasurface-Inspired ISRJ with Gate Reconstruction

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Abstract—Classical intermittent sampling repeater jamming (ISRJ) parameter estimation usually assumes ideal rectangular gating. This assumption is insufficient for metasurface-inspired ISRJ, where the signal may retain a finite low-scattering state instead of switching fully off. This paper studies the estimation of slice width, sampling interval, and the low-scattering coefficient directly from raw IQ observations. A Transformer-based estimator is used to capture the global periodic structure of the ISRJ signal. A one-period gate-reconstruction branch is added as an auxiliary structural constraint to help the model follow the modulation pattern. On the current synthetic benchmark, the proposed method reaches a total measurement accuracy of 83.61%, and the total measurement accuracy increases to 95.62% when JNR is above 0 dB. The results show that the proposed model is effective for this parameter-estimation task, while low-JNR robustness remains the main challenge.

Index Terms—radar countermeasure, interrupted sampling, metasurface-inspired deception, parameter estimation, trans-former

I. INTRODUCTION

Intermittent sampling repeater jamming (ISRJ) is an important form of coherent repeater jamming. The jammer intercepts the radar signal, samples it in short slices, and retransmits the samples after delay [1]. This sampling-and-retransmission process can generate multiple false targets while avoiding full-pulse storage [2]. The timing law is therefore central to both jamming analysis and countermeasure design. For this reason, classical ISRJ studies have focused mainly on slice width, sampling interval, and the resulting false-target structure [3], [4].

Deep learning has recently been introduced into radar jamming analysis and parameter inference. Wang *et al.* [5] estimate ISRJ parameters through an end-to-end pipeline that combines separation and measurement. Lu *et al.* [6] also couple recognition and parameter estimation in a detection-style framework. These studies show that data-driven models can reduce the dependence on handcrafted processing. However, they are still built around conventional ISRJ assumptions.

In metasurface-inspired ISRJ, the returned signal is governed by switching between high- and low-scattering states. Sun *et al.* [7] describe how time-modulated metasurfaces (TMMs) use periodic switching between reflection states to manipulate radar echoes. Ding *et al.* [8] further show that TMMs can be used in radar jamming applications. Cai and Li [9] extend this picture by using simultaneous phase and amplitude coding to realize multi-frequency electromagnetic

modulation, which supports a two-state view with a nonzero low-scattering level. Existing ISRJ parameter-estimation studies do not directly address this case, where the signal departs from ideal rectangular gating but still retains periodic structure.

This paper studies parameter estimation for metasurface-inspired ISRJ with a normalized high-scattering state and a nonzero low-scattering state. The goal is to estimate slice width, sampling interval, and the amplitude coefficient of the low-scattering state from raw IQ observations. These parameters are all governed by the same periodic gate pattern. A direct three-output regressor does not enforce that structure explicitly. This paper therefore uses a reconstruction-guided estimator that jointly estimates the three parameters while recovering the gate profile, thereby tying the estimation process to the waveform structure of the ISRJ signal. Its main contributions are:

- 1) A reconstruction-guided estimator that jointly estimates slice width, sampling interval, and the amplitude coefficient of the low-scattering state while recovering the underlying periodic modulation waveform.
- 2) Strong results on the current benchmark, including high total measurement accuracy when JNR is above 0 dB.

The remainder of the paper defines the signal model, presents the proposed method, and reports the experimental results.

II. SIGNAL MODEL AND PROBLEM FORMULATION

In this paper, metasurface-inspired ISRJ is modeled as periodic switching between a high-scattering state and a low-scattering state. This formulation bridges idealized ISRJ gating and metasurface-inspired two-state modulation while remaining simple enough for parameter estimation. A linear-frequency-modulated waveform is used as the base signal, and the target delay and deceptive delay are denoted by τ_t and τ_j , respectively.

After normalizing the high-scattering state to one and denoting the amplitude coefficient of the low-scattering state by $x_{\text{low}} \in [0, 0.5]$, the modulation waveform is written as

$$p(t) = (1 - x_{\text{low}}) \text{rect}\left(\frac{t - \tau_j}{T_l}\right) \otimes \sum_{k=-\infty}^{+\infty} \delta(t - \tau_j - kT_s) + x_{\text{low}}, \quad (1)$$

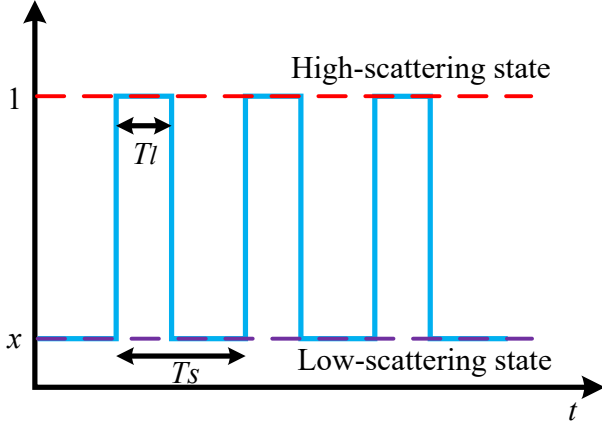


Fig. 1. Time modulation model with alternating high- and low-scattering states.

where $\text{rect}(\cdot)$ denotes the rectangular pulse, $\otimes$ denotes convolution, T_l is the slice width, and T_s is the sampling interval. Under this model, the effective scattering coefficient switches periodically between a normalized high-scattering state of 1 and a low-scattering state of x_{low} .

The metasurface-inspired ISRJ signal is then expressed as

$$r_j(t) = p(t) s(t - \tau_j), \quad (2)$$

and the received signal becomes

$$r(t) = s(t - \tau_t) + r_j(t) + n(t), \quad (3)$$

where $n(t)$ is complex Gaussian noise. The parameter-estimation target is the vector

$$\mathbf{y} = [T_l, T_s, x_{\text{low}}]^\top. \quad (4)$$

One derived quantity is used later in the decoder:

$$D = \frac{T_l}{T_s}, \quad (5)$$

which gives the fraction of each period spent in the high-scattering state.

III. PROPOSED METHOD

The proposed method estimates the slice width T_l , sampling interval T_s , and the amplitude coefficient of the low-scattering state x_{low} directly from the received IQ signal. Rather than treating these three quantities as independent regression targets, the model predicts them through an intermediate gate-reconstruction pathway. This design follows the signal model in Section II: all three parameters are governed by the same periodic switching pattern. Reconstructing that pattern therefore provides a structural constraint that keeps the estimator consistent with the underlying jammer physics.

A. Input Representation

The estimator uses a two-channel IQ input. Let $r[n] \in \mathbb{C}$ denote the sampled received signal. Its real and imaginary parts are stacked as

$$\mathbf{X} = \begin{bmatrix} \Re\{r[0]\} & \cdots & \Re\{r[N-1]\} \\ \Im\{r[0]\} & \cdots & \Im\{r[N-1]\} \end{bmatrix} \in \mathbb{R}^{2 \times N}. \quad (6)$$

B. Encoder

Figure 2 shows the proposed estimator in two parts: Fig. 2(a) illustrates the encoder, and Fig. 2(b) illustrates the decoder. The encoder begins with a convolutional tokenizer, implemented as a shallow temporal convolutional front-end followed by a strided patch-embedding layer, which converts the received IQ sequence into latent patch representations. A learnable global embedding is concatenated with the patch sequence, positional encoding is added, and the resulting sequence is processed by a transformer encoder.

The output sequence is then separated into a global embedding and patch-level contextualized representations. The patch sequence is summarized by mean pooling, max pooling, and standard-deviation pooling across the sequence dimension. These three pooled vectors are concatenated with the global embedding to form a shared summary vector,

$$\mathbf{h}_{\text{sum}} = [\mathbf{h}_g; \text{Mean}(\mathbf{H}); \text{Max}(\mathbf{H}); \text{Std}(\mathbf{H})], \quad (7)$$

where $\mathbf{h}_g$ denotes the global embedding and $\mathbf{H}$ denotes the matrix of patch-level contextualized representations. This summary vector serves as the common feature source for all downstream predictions.

C. Decoder

As shown in Fig. 2(b), the decoder first splits the encoder output and aggregates it into the shared summary $\mathbf{h}_{\text{sum}}$. The first decoder branch maps $\mathbf{h}_{\text{sum}}$ to a timing representation through a multilayer perceptron and then predicts the sampling interval $\hat{T}_s$. The resulting timing representation is concatenated back with the shared summary so that the remaining branches are conditioned on the estimated timing context.

The second branch predicts the amplitude coefficient of the low-scattering state $\hat{x}_{\text{low}}$. The third branch predicts a discretized gate profile over one switching period,

$$\hat{\mathbf{g}} = [\hat{g}_1, \hat{g}_2, \dots, \hat{g}_K]^\top, \quad (8)$$

where K is the number of gate bins. Each element represents the predicted scattering level at one normalized position within the switching period. This profile induces a soft duty ratio,

$$\hat{D} = \frac{1}{K} \sum_{k=1}^K \hat{g}_k, \quad (9)$$

from which the slice width is recovered jointly with the predicted sampling interval as

$$\hat{T}_l = \hat{D} \hat{T}_s. \quad (10)$$

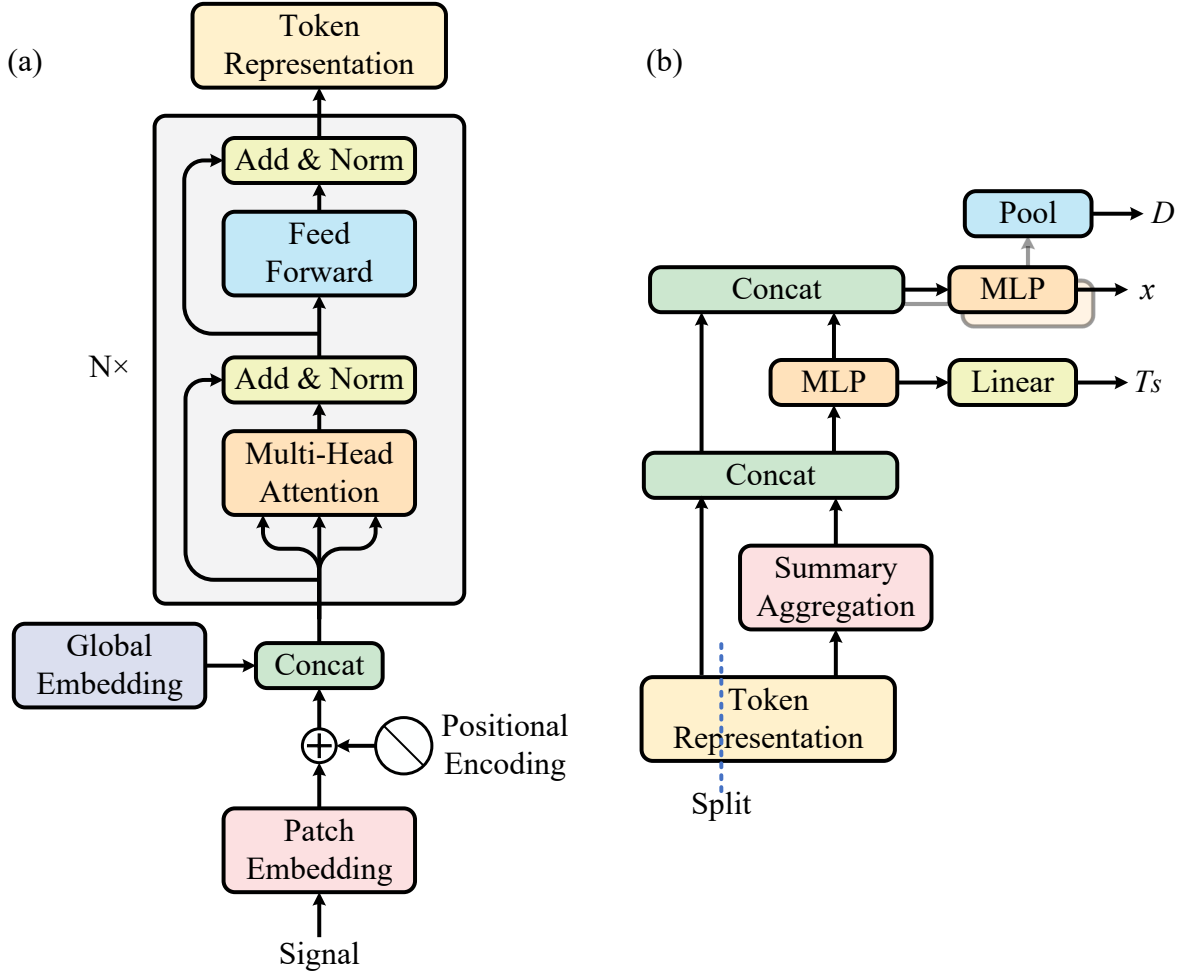


Fig. 2. Architecture of the proposed gate-reconstruction estimator. (a) The encoder converts the input IQ sequence into token representations by combining patch embedding, a learnable global embedding, positional encoding, and transformer encoding. (b) The decoder splits the token representations, aggregates them into a shared summary, and predicts the sampling interval T_s , the amplitude coefficient x of the low-scattering state, and the duty ratio D .

The decoded one-period profile is then mapped back to the full time axis to form the jammer mask

$$\hat{\mathbf{m}} = [\hat{m}_1, \hat{m}_2, \dots, \hat{m}_L]^\top, \quad (11)$$

where each $\hat{m}_n$ is obtained by repeating the one-period profile according to the predicted interval $\hat{T}_s$ and combining it with the predicted amplitude coefficient of the low-scattering state $\hat{x}_{\text{low}}$. The predicted timing parameters are clipped to satisfy the dataset constraints, particularly the minimum separation between adjacent high-scattering segments. This step ensures that the decoded gate remains physically realizable.

D. Model Optimization

Let $\hat{\mathbf{y}} = [\hat{T}_l, \hat{T}_s, \hat{x}_{\text{low}}]^\top$ denote the decoded physical predictions, and let $\hat{\mathbf{m}}$ denote the reconstructed full-length jammer mask. The training objective combines parameter regression, mask reconstruction, and total-variation (TV) regularization:

$$\mathcal{L} = \mathcal{L}_{\text{reg}} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}}, \quad (12)$$

where the parameter regression term is

$$\begin{aligned} \mathcal{L}_{\text{reg}} = & w_l \ell_{\text{SL1}}(\hat{T}_l, T_l) \\ & + w_s \ell_{\text{SL1}}(\hat{T}_s, T_s) + w_x \ell_{\text{SL1}}(\hat{x}_{\text{low}}, x_{\text{low}}), \end{aligned} \quad (13)$$

the mask-reconstruction term is

$$\mathcal{L}_{\text{mask}} = \ell_{\text{SL1}}(\hat{\mathbf{m}}, \mathbf{m}), \quad (14)$$

and the TV regularizer on the predicted one-period gate profile is

$$\mathcal{L}_{\text{tv}} = \frac{1}{K-1} \sum_{k=1}^{K-1} |\hat{g}_{k+1} - \hat{g}_k|. \quad (15)$$

Here, $\ell_{\text{SL1}}(\cdot, \cdot)$ denotes the smooth- L_1 loss,

$$\ell_{\text{SL1}}(a, b) = \begin{cases} \frac{1}{2}(a-b)^2, & |a-b| < 1, \\ |a-b| - \frac{1}{2}, & |a-b| \geq 1, \end{cases} \quad (16)$$

The regression term supervises the final physical parameters, the mask term aligns the reconstructed waveform-level modulation with the ground-truth jammer mask, and the TV term

TABLE I
PARAMETER RANGES USED IN DATA GENERATION.

Symbol	Range
T_l	0.4~4.0 μs
T_s	1.0~10.0 μs
x	0.0~0.5
SNR	0 dB
JNR	-10~20 dB

suppresses unrealistic bin-to-bin oscillations in the predicted period profile.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

1) *Dataset Settings*: The dataset is generated according to the signal model in Section II. The ranges of T_l , T_s , x , and JNR are listed in Table I, while the SNR is fixed at 0 dB. For each sample, the signal parameters are randomly drawn from these ranges to generate different metasurface-inspired ISRJ signals. The resulting dataset contains 6200 training samples, 775 validation samples, and 775 test samples.

2) *Training Settings*: The network is optimized with a learning rate of 3×10^{-4} and a weight decay of 10^{-4} .

B. Evaluation Metrics

Both mean absolute error (MAE) and measurement accuracy are reported so that the evaluation reflects both average deviation and operational tolerances. Let N_{test} denote the number of test samples. Let $\hat{T}_{l,i}$, $\hat{T}_{s,i}$, and $\hat{x}_{\text{low},i}$ denote the predicted parameters of sample i , and let $T_{l,i}$, $T_{s,i}$, and $x_{\text{low},i}$ denote the corresponding ground-truth values. The three per-parameter measurement accuracy metrics are defined as

$$A_{T_l} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(|\hat{T}_{l,i} - T_{l,i}| \leq 0.15 \mu s) \times 100\%, \quad (17)$$

$$A_{T_s} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(|\hat{T}_{s,i} - T_{s,i}| \leq 0.25 \mu s) \times 100\%, \quad (18)$$

$$A_x = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(|\hat{x}_{\text{low},i} - x_{\text{low},i}| \leq 0.05) \times 100\%, \quad (19)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Let $\mathcal{E}_i$ denote the event that all three parameter errors of sample i satisfy their respective tolerances. The total measurement accuracy is then defined as

$$A_{\text{total}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(\mathcal{E}_i) \times 100\%. \quad (20)$$

C. Overall Results

Table II summarizes the test results over the five runs. The proposed method reaches a total measurement accuracy of $83.85\% \pm 1.10\%$, indicating that all three parameters satisfy their prescribed tolerances for most test samples.

For comparison, a 1D residual convolutional baseline was used. It uses global average pooling and a three-parameter

TABLE II
OVERALL PARAMETER RECOVERY RESULTS.

Method	Parameter	MAE	Measurement Accuracy (%)
Proposed	T_l	0.0588 ± 0.0020	93.52 ± 0.39
	T_s	0.0995 ± 0.0066	95.79 ± 0.68
	x_{low}	0.0265 ± 0.0014	87.72 ± 0.98
	Total	—	83.85 ± 1.10
ResNet	T_l	0.1084 ± 0.0029	81.55 ± 1.79
	T_s	0.2308 ± 0.0083	73.24 ± 1.15
	x_{low}	0.0412 ± 0.0033	70.12 ± 4.51
	Total	—	50.97 ± 3.13

regression head. This baseline was trained with the same IQ input and the same optimization settings as the proposed method. As shown in Table II, it reaches a total measurement accuracy of $50.97\% \pm 3.13\%$, which is far below the $83.85\% \pm 1.10\%$ achieved by the proposed method. The gap is even larger in the low-JNR regime, where it obtains only $22.48\% \pm 2.20\%$ total measurement accuracy. These results show a clear advantage of the proposed framework over a direct convolutional baseline.

Looking at the individual parameters, the results show a clear pattern. Slice width and sampling interval are easier to recover, with measurement accuracy of $93.52\% \pm 0.39\%$ and $95.79\% \pm 0.68\%$, respectively, while the amplitude coefficient of the low-scattering state remains harder, with a measurement accuracy of $87.72\% \pm 0.98\%$. The same trend appears in the mean absolute errors: both T_l and T_s remain below $0.1 \mu s$ on average, whereas the amplitude coefficient of the low-scattering state remains the dominant residual error source.

D. Results Across JNR Conditions

Fig. 3 shows the parameter measurement accuracy under different JNR conditions. All four curves improve as JNR increases, and the strongest degradation is concentrated in the amplitude coefficient of the low-scattering state and in the total measurement accuracy.

This behavior is consistent with the signal model. The timing parameters T_l and T_s mainly determine the periodic structure of the gate, so they are supported by global cues such as repetition pattern and edge location across the waveform. By contrast, x mainly controls the amplitude contrast between the high and low states through $p(t) = x + (1 - x)g(t)$, which provides weaker structural evidence. In addition, the information about x is concentrated mainly in the low-state segments, while the high-state segments are normalized to unity and therefore contribute less to its identification. As a result, the estimation of x is more sensitive to noise contamination, which explains why its measurement accuracy degrades more severely in the low-JNR regime.

Table III further summarizes the low-, mid-, and high-JNR ranges. In the low-JNR range $[-10, 0)$ dB, the total measurement accuracy drops to $57.12\% \pm 2.49\%$, and the measurement accuracy of the amplitude coefficient of the low-scattering state falls to $67.60\% \pm 2.10\%$. In contrast, the model is already close to saturation in the medium- and

TABLE III
PARAMETER MEASUREMENT ACCURACY IN THE LOW-, MID-, AND
HIGH-JNR RANGES.

JNR range (dB)	A_{T_i} (%)	A_{T_s} (%)	A_x (%)	A_{total} (%)
$[-10, 0)$	81.44 ± 0.88	87.20 ± 2.08	67.60 ± 2.10	57.12 ± 2.49
$[0, 10)$	99.20 ± 0.40	99.76 ± 0.22	95.60 ± 1.62	94.80 ± 1.62
$[10, 20]$	99.35 ± 0.70	100.00 ± 0.00	98.84 ± 0.75	98.18 ± 1.23

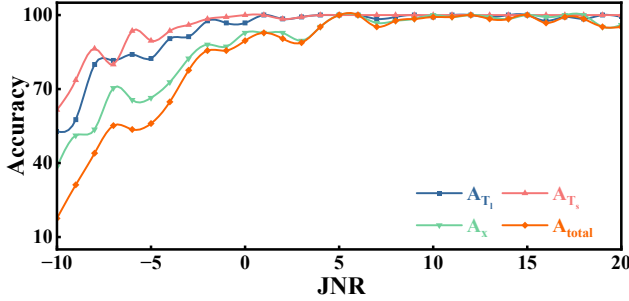


Fig. 3. Parameter measurement accuracy under different JNR conditions.

high-JNR ranges, where both timing parameters exceed 99% measurement accuracy on average.

V. CONCLUSION

This paper presented a gate-reconstruction learning framework for parameter estimation under a metasurface-inspired interrupted-sampling deception model with a nonzero amplitude coefficient of the low-scattering state. The main idea is to couple parameter decoding with periodic gate recovery instead of treating the task as three independent scalar regressions.

On the synthetic benchmark used in this paper, the method reaches a mean total measurement accuracy of $83.85\% \pm 1.10\%$ on the test split over five independent runs. Performance is already near-saturated in moderate- and high-JNR regimes, while robustness in extremely low JNR conditions remains the main unresolved weakness.

These findings support the central claim of the paper: under the considered two-state ISRJ model, reconstructing the gate profile helps the model estimate slice width, sampling interval, and the amplitude coefficient of the low-scattering state more reliably. The present study is limited to a simulation-based single-component setting with a narrow jammer family and one heuristic baseline. The claims should therefore be read within that scope. Future work should strengthen low-JNR robustness, extend the benchmark to broader jammer families, and validate the framework beyond the current single-component synthetic setting.

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